CENG-322 Mobile Application Development - Deliverable 1

Team Name: IoT Innovators

Project Name: Smart Farm

Team Number: 8

Student's names and IDs:

- Chau Vi Cuong Nguyen – N01634611
- Hoang Tran – N01613506
- Tuan Anh Pham – N01238459
- Truong Anh Nguyen - N01639800

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Team Contract (from CENG-322_TEAM_contract_v2.0)

CENG-322 TEAM PROJECT

Team Name: IoT Innovators_____

Team Number: _____8_____

Project Name: ____Smart Farm_____

Please negotiate, sign, scan and include as the first section in your Deliverable 1.

Please note that if cheating is discovered in a group assignment each member will be charged with a cheating offense regardless of their involvement in the offense. Each member will receive the appropriate sanction based on their individual academic honesty history.

Please ensure that you understand the importance of academic honesty. Each member of the group is responsible to ensure the academic integrity of all of the submitted work, not just their own part. Placing your name on a submission indicates that you take responsibility for its content.

Team Member Names (Please Print)	Signatures	Student ID	Github Id
Chau Vi Cuong Nguyen	Cuong	N01634611	ChauViCuongNguyen4611
Hoang Tran	Tran	N01613506	HoangTranTran3506
Tuan Anh Pham	Anh	N01238459	TuanAnhPham8459

Truong An Nguyen	An	N01639800	TruongAnNguyen9800

For further information read Academic Honesty Policy on https://humber.ca/legal-and-risk-management/policies/search-by-students.html.

By signing this contract, we acknowledge having read the Humber Academic Honesty Policy as per the link below.

https://academic-regulations.humber.ca/2018-2019/17.0-ACADEMIC-MISCONDUCT

Responsibilities of the Project Leader include:

- Assigning tasks to other team members, including self, in a fair and equitable manner.
- Ensuring work is completed with accuracy, completeness and timeliness.
- Planning for task completion to ensure timelines are met
- Any other duties as deemed necessary for project completion

What we will do if . . .

Scenario	**Accepted initials**	**We agree to do the following**
Team member does not deliver component on time due to severe illness or extreme personal problem	CVCN HT TA TAN	**a)** Team absorbs workload temporarily _x_ d) Other:
Team member cannot deliver component on time due to lack of ability	CVCN HT TA TAN	a) Team reassigns component __ **b)** Team helps member _x_ c) Team "fires" team member by not permitting his/her name on submission __ d) Other:

Team member does not deliver component on time due to lack of effort	CVCN HT TA TAN	a) Team absorbs workload __ b) Team "fires" team member by not permitting his/her name on submission __ **c) Other**: Team absorbs workload and change the mission for that team member
Team member does not attend team meeting	CVCN HT TA TAN	a) Team proceeds without him/her and will assign work to the absent member __ b) Team doesn't proceed and records team member's absence __

		c) Team proceeds for that meeting but "fires" member after _3_ occurrences _x_
An unforeseen constraint occurs after the deliverable has been allocated and scheduled (a surprise test or assignment)	CVCN HT TA TAN	a) Team meets and reschedules deliverable _x_ b) Team will cope with constraint __ c) Other:
Team cannot achieve consensus leaving one member feeling "railroaded", "ignored", or "frustrated" with a decision which affects all parties	CVCN HT TA TAN	a) Team agrees to abide by majority vote _x_ b) Team flips coin __ c) Other:

Team members do not share expectations for grade desired	CVCN HT TA TAN	a) Team will elect one person as "standards-bearer" who has the right to ask that work be redone __ b) Team votes on each submission's quality __ c) Team will ask for individual marking and will identify sections by author _x_ d) Other:
Team member behaves in an unprofessional manner by being rude or uncooperative	CVCN HT TA TAN	a) Team attempts to resolve the issue by airing the problem at team meeting __ b) Team ignores behaviour __ c) Team agrees to avoid use of all vocabulary inappropriate to the business setting _x_ d) Team fires the team member.
Team member assumes or requests that his/her name be signed to a submission but has not participated in production of the deliverable	CVCN HT TA TAN	a) Team agrees that this is cheating and is unethical _x_ b) Friends are friends and should help each other __

		c) That person name will not be put on the submission._
There is a dominant team member who is content to make all decisions on the team's behalf leaving some team members feeling like subordinates rather than equal members	CVCN HT TA TAN	a) Team will actively solicit consensus on all decisions which affect project direction by asking for each member's decision and vote __ b) Team will express subordination feelings and attempt to resolve issue __ **c)** Other: Team will have a vote, and decision will be made based on majority
Team has a member who refuses to participate in decision making but complains to others that s/he wasn't consulted	CVCN HT TA TAN	a) Team forces decision sharing by routinely voting on all issues __ b) Team routinely checks with each other about perceived roles __

		c) Team discusses the matter at team meeting _x_

Signature of each student who participated in the work (team members who did not participate and where their signature is missing will get 0)

CVCN

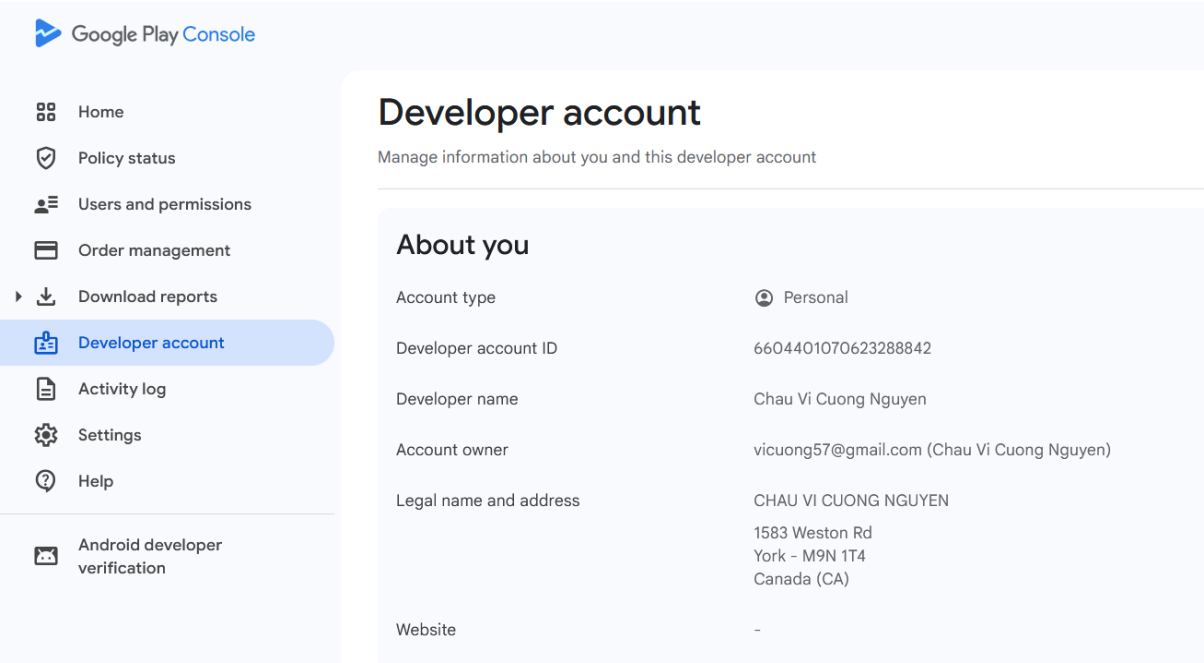
HT

TA

TAN

Create Google Play Developer Account

Take a screenshot showing account has been approved and created (one or more members).

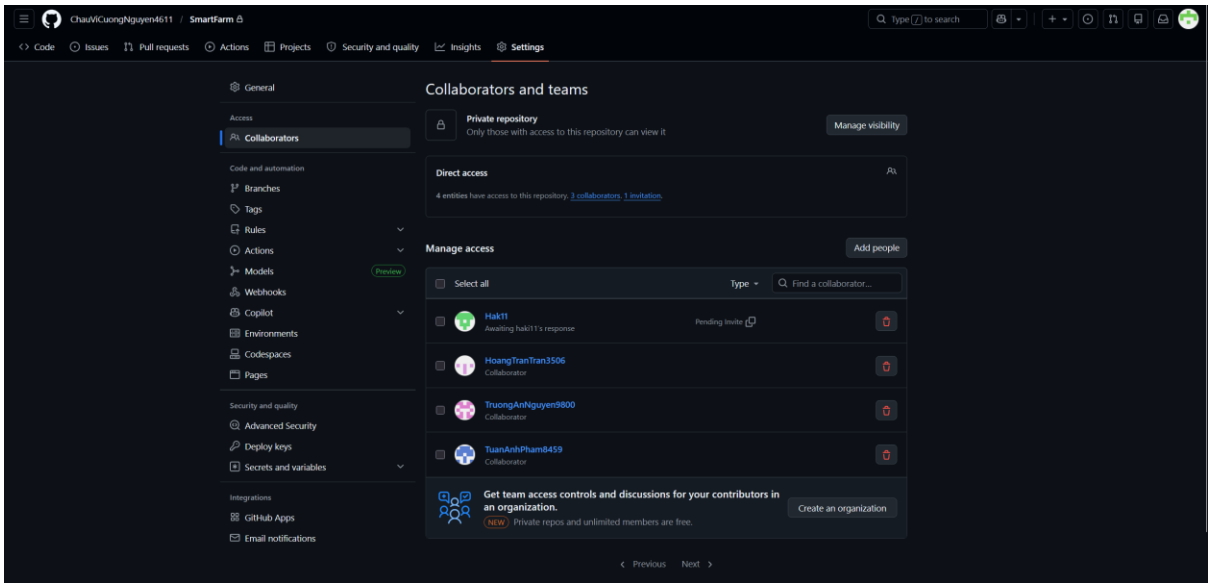


Link GitHub Repo.

All members must contribute to the repo.

https://github.com/ChauViCuongNguyen4611/SmartFarm

Screenshot of github invitation.



Project Background and Description:

1. Describe the project goals and final vision:

The Smart Farm IoT application aims to create a comprehensive mobile solution for modern farming management. The final vision is to provide farmers with real-time monitoring and control of IoT-enabled farm equipment, including soil moisture sensors, temperature monitors, automated irrigation systems, and crop health trackers.

2. Describe the software aspect and hardware.

Software: Android application using Java, Firebase Cloud Database, Material Design Components

Hardware: IoT sensors (soil moisture, temperature, humidity), microcontrollers, actuators for irrigation systems

3. Describe the screen flow.

- User taps on app's homescreen icon, the splash screen appear with some animations for 3 seconds
- System Dashboard (Home) will be displayed by default
- User can click on menu icon on top-right corner to open navigation drawer and to go through different fragments.

4.Demonstrate how you are planning to satisfy to read / write from the DB which is hosted on the cloud.

The app will utilize Firebase SDK for reading real-time data updates from the database. It will also save numbers of latest data on local storage for offline mode.

5.How you Incorporated the feedback provided through the interview.

Added offline mode support based on farmer connectivity concerns

Simplified UI for easier field use

Added emergency alerts for critical farm conditions.

Project Scope:

Describe the technical scope of the project by talking about the project plan, and how you will know when the project is complete.

The project involves designing and developing a Smart Farm that monitors environmental conditions and allows users to manage climate settings through an Android application. The system consists of hardware sensors connected to a Raspberry Pi, which collects data such as temperature, humidity, soil moisture, and light levels. The Raspberry Pi processes the sensor data and communicates with the Android application using a network connection.

The project plan is divided into several phases. The first phase focuses on researching suitable sensors, configuring the Raspberry Pi, and establishing communication between the hardware and software components. The second phase involves developing the Android application, including the dashboard, climate control panel, analytics screens, and settings interface. The third phase focuses on integrating hardware and software systems, implementing real-time data updates, and testing communication between the Raspberry Pi and the mobile application.

The final phase includes system testing, debugging, performance improvements, and user interface refinements.

The project will be considered complete when all planned features are successfully implemented and tested. Completion criteria include accurate sensor data collection, successful transmission of data to the Android application, functional climate control features, real-time monitoring capabilities, analytics and reporting functions, configurable settings, and a stable user interface. The system must also demonstrate reliable operation during testing, with sensor readings updating correctly and user commands being executed as expected.

Describe how to integrate the software with the hardware components.

The data from sensors will be read using mainly I2C interface on Raspberry.

It will then be processed and uploaded to cloud database.

Real-time listeners in Android app will detect and calculate new data to match user's settings and preferences.

Project layout:

Write how you plan to implement Navigation Drawer, and combination with another layout.

How to use them, why to use, use cases of existing applications that utilize them. one vs the others, and use cases if existing apps that use the above.

- Navigation Drawer slides from the left edge, accessed by tapping the hamburger icon ☰ or swiping from left.
- Implementation plan
  - We will implement a Navigation Drawer (hamburger menu) using Android's Navigation View inside a Linear Layout.
  - Combination with another layout:
    The main content swaps Fragment objects (Dashboard, History, Settings, Profile).
- Why use Navigation Drawer
  - Standard Android pattern (familiar to users).

- Allows quick switching between top-level sections without losing context.

Write Minimum of 1 theme:

For the Theme write:

Two epics, 3 stories for each epic, 3 tasks for each story.

Theme: Smart farm animal monitoring and automation system.

Epic 1: Environmental monitoring and control.

- Story 1: As a farmer, I want to monitor the temperature and humidity of my farm in real time so that I can keep the plants in a safe environment.

+ Tasks:

- Connect to the temperature and humidity.
- Send live sensor data to the app.
- Test if the readings are updated correctly.

- Story 2: As a farmer, I want the system to monitor the soil moisture of my farm so that I can know when exactly I need to water them.

+ Tasks:

- Integrate soil moisture sensors into the system architecture.
- Display the current soil moisture data on the app.
- Calibrate the sensors to ensure accurate wet/dry reading.

- Story 3: As a farmer, I want the system to automatically trigger irrigation and fans when the conditions meet so that I can save time and resources.

+ Tasks:

- Program threshold values for minimum soil moisture and maximum temperature.
- Connect the microcontroller to water pumps and ventilation fans.
- Run tests to ensure pumps and fans turn on automatically when conditions are met.

Epic 2: System management and user notifications.

- Story 1: As a farmer, I want to monitor lighting conditions so that I can ensure the plants receive the most optimal amount of light for healthy growth.

+ Tasks:

- Integrate light intensity sensors (e.g., photoresistor or digital light sensor) into the system.
- Display real-time light levels (lux or percentage) on the app dashboard.
- Set day/night cycle detection and alert the farmer if light levels fall below a configurable threshold.

- Story 2 :
  As a farmer, I want to receive mobile notifications when environmental conditions go out of range so that I can take quick action even when away from the farm.

+ Tasks:

- Implement push notification service (e.g., Firebase Cloud Messaging or Blynk).
- Trigger notifications for high temperature, low soil moisture, or abnormal light levels.
- Allow farmers to enable/disable specific notification types in the app.

- Story 3 :
  As a farm owner, I want to view historical sensor data trends so that I can make better long-term decisions about irrigation and climate control.

+ Tasks:

- Store sensor readings (temp, humidity, soil moisture, light) in a time-series database.
- Create charts showing daily/weekly/monthly trends in the app.
- Provide an export option (CSV) for manual analysis.